

Dream-RSI Methods and Results

A construction log for long, genuinely readable English letter-level palindromes | Snapshot: 2026-09-18

Current status: no reader-worthy output has cleared the full gate yet. This report is intentionally not a negative-result thesis: it records the working search machinery, the exact diagnostic surfaces, and the next geometry-changing repair required to reach a long readable result.

1. Acceptance gate

A candidate is admitted only when (1) normalized letters equal their reverse, verified by an independent two-pointer check and an independent forward/reverse SHA-256 check; (2) provenance shows fresh construction rather than catalogue text, finished-tape reversal, word-order symmetry, repeated units, or self-palindromic words/spans; (3) the surface is intact ordinary English prose; and (4) blinded human readers prefer the intact passage to a randomized word-shuffled control. Programmatic language measures may filter or diagnose, but never certify readability.

2. Dream-RSI controller

Each completed construction lane writes a replay node: rendered surface, normalized tape, exactness hashes, mismatch mask, grammar and semantic state, provenance flags, and a concrete next repair. Replay selects a new operator on a held-out split; a fresh branch then executes that operator. A failure is progress only when it changes the construction geometry and names the next repair. The controller is an orchestration layer, never a readability judge.

The current live operator freezes matched outer character assignments, reopens the seam-owning typed constituents, and lets lexical content, inflection, and word boundaries change while carrying the mirrored character obligation immediately. The latest branch also carries subject/verb/object/adjunct role labels and clause feasibility through each character expansion; its next repair replaces the sparse corpus chart with a typed hand-authored clause automaton.

3. Actual Dream-RSI results

Round / lane	Fresh work	Exact / admitted	Reader status
Model-guided span resynthesis	12 local GPT-2 seam proposals; exact rows were constructed by mirroring the residual; 86-120 letters.	12 exact by construction / 0 shortcut-free	Human-unreviewed; reflected residual had no lexical segmentation.
Replay round 119	21,990 independently audited nodes; 120 exact-but-fragment controls entered replay.	0 admissible exact	No reader gate; held-out mismatch-first stayed 0.780.
Dual-boundary model authoring	8 left x 8 right model proposals; 64 fresh paired worlds; no seed scaffold in output.	0 exact; longest 58 letters	Human-unreviewed; next repair reopens a complete residual word.
Discourse ellipsis + cross-word seam	148 exact diagnostic rows, 42-46 letters; 100 passed word-order check, but all retained rows have a forbidden proper palindromic island.	148 exact / 0 strict	Reader gate closed; next repair jointly authors a complete answer clause.
Role-live phrase lattice	34 fresh model units (1 subject, 11 verb phrases, 11 objects, 11 adjuncts); role order and clause prefix/suffix feasibility were live constraints.	0 exact; 189,481 states rejected	Reader gate closed; Brown shape inventory is too sparse; next repair is a typed hand-authored clause automaton.
Seed-extension wrapper search	25 fresh independently authored prefix/suffix wrapper pairs around an internal seed state; outer equations were live before rendering.	0 exact; longest 103 letters	Reader gate closed; next repair authors held-out wrappers ending in the first residual character class.

Round / lane	Fresh work	Exact / admitted	Reader status
Reader-first complete-clause lattice	10 human-authored complete scenes with explicit subject, verb, object, and adjunct roles; 400 typed center/edge lattice states.	0 exact; longest 67 letters	Reader gate closed; catalogue preflight still requires an explicit rerun; next repair carries residuals through typed verb-object transitions.
Character-trie provenance correction	The apparent 51-letter result was three hard-coded published palindromes, retained only as regression controls after audit.	0 fresh / 0 admissible	Quarantined; next repair must use a genuinely fresh held-out phrase inventory.
Fresh phrase-trie product	232,320 fresh complete-clause records; live obligations walked 947,222 reversed-right trie nodes before rendering any pair.	0 terminal pairs / 0 exact	Reader gate closed; next repair adds a center-state grammar transition and held-out semantic valency frames.
Center-state grammar transition	2,027,520 live trie walks with a typed observation-to-consequence center transition; 48 fresh diagnostics were rendered.	0 terminal pairs / 0 exact	Reader gate closed; next repair carries held-out semantic valency agreement across the center.
Held-out semantic valency frames	Six complete authored subject/verb/object/adjunct frame pairs with agreement and valency live before rendering; longest pair 72 letters.	0 exact	Reader gate closed; next repair adds a held-out adjunct transition at the center seam.
Fresh reverse-tape grammar chart	Six fresh authored clauses consumed by a typed DET/ADJ/N/V/P chart with optional-slot transitions; best intact control 42 letters.	0 exact / 0 shortcut survivors	Reader gate closed; next repair adds typed plural and inflectional variants at the first residual character.

The strongest exact diagnostic surface

Independent audit: 44 normalized letters; two-pointer exact = true; forward SHA-256 = reverse SHA-256 = 0419d58b6773dcca6d527271d41fd0197da9dde876598c9dadfc87fa4f1a9f83. It is not admitted: the surface contains a proper palindromic island and therefore fails the shortcut-free gate. It has not been sent to readers.

Was Noel an era, a gas, an item? Met in a, saga, arena, Leon saw.

This row is useful because it demonstrates genuine cross-word boundary resegmentation (an era / arena, a gas / saga, an item / met in a) while also exposing the remaining failure mode: an exact tape can still be an engineered word-mirror or contain a proper palindrome island. The construction must now generate the whole discourse clause jointly and reject every such subspan before closure.

4. Fifty distinct construction methods and results

The matrix below is extracted from the novelty registry. Each row is a distinct construction signature, not a repeated parameter sweep. Results preserve the registry's wording; entries marked preflight have no run result and are not evidence of success. Exact diagnostic rows remain quarantined unless every shortcut-free gate passes.

#	Method / operator	Construction signature	Observed result and next repair
1	phrase boundary valency equations	typed-transitive-frames / slot-boundary-character-equations / pre-render-propagation / independent-two-pointer-sha-audit	diagnostic run; 1,562,500 independent frame-pair states, all pruned by pre-render equations, 0 exact closures; intact controls 49–53 letters; reader gate closed; next: optional determiner/adjunct boundary carry with held-out frames
2	semordnilap crossword resegmentation	fresh-scene-left-arm / independent-semordnilap-word-pairs / cross-word-boundary-shifts / typed-clause-realization / two-pointer-sha-audit / no-repeated-uni	diagnostic run; 150 crossword boundary probes, 0 exact closures, longest 55 letters; independent SHA audit rejects all rows; next: first-residual typed lexical expansion
3	cross attachment resegmentation	cross-attachment-character-carry / determiner-adjunct-resegmentation / typed-valency / independent-two-pointer-hash-audit	diagnostic run; 0 equation nodes and 0 exact closures; intact controls 58-70 letters; reader gate closed; next: typed adjunct insertion with boundary carry

#	Method / operator	Construction signature	Observed result and next repair
4	typed morpheme compound boundary	typed-prefix-stem-suffix / compound-boundary / independent-ordinary-clauses / inflectional-derivational-variants / independent-pointer-sha / no-word-mirror	diagnostic run; 54 probes, no exact closure, longest 59 letters
5	adjunct boundary targeted repair	single-adjunct-boundary-repair / fresh-preposition / preserved-typed-svo / live-seam-residual / independent-audit	repair run; exact closure unresolved
6	fresh typed frame live seam	fresh-typed-svo-frame / agreement-preserved / live-mirrored-character-obligation / seam-residual-ledger / independent-audit	repair run; residual obligations remain
7	clause pair csp central pivot	typed-svo-valency / independent-scene-pairs / central-pivot / character-tape-csp / no-word-mirror	diagnostic run; 113-letter intact prose, unresolved character debt
8	semantic slot lattice smt	semantic-slot-lattice / global-character-equation-pruning / joint-left-right-selection / weighted-residual-repair / independent-audit	diagnostic run; 108 rendered states, 0 accepted
9	bidirectional scene decoder	bidirectional-character-lm / live-semantic-scene-slots / outside-in-tape / independent-audit	diagnostic run; exact catalogue palindrome rejected
10	wordpair graph	independent-word-pair-graph / pos-valency-state / distinct-word-pairs / character-tape-closure / independent-pointer-hash	diagnostic run; 164-letter intact prose frontier, 0 exact closures
11	paired semantic mutation	quarantined-control / paired-mirrored-lexical-span / scene-slot-grammar / live-character-obligations	diagnostic run; 2 rendered candidates, 0 exact and 0 admitted
12	corpus backed reverse segmentation	fresh-clause-tape / weighted-word-boundary-dp / independent-pointer-hash / copied-span-rejection	diagnostic run; 2 fresh clauses, 0 exact closures
13	seam feature slot repair	targeted-seam-feature-slot-repair / single-lexical-substitution / live-reflected-seam / independent-exact-audit	repair run; substitutions did not close exact seam
14	typed boundary resegment shortwords	typed-first-failing-seam / boundary-resegmentation / adjacent-short-word-absorption / fresh-prose-control / independent-pointer-sha-audit	repair run; 23 variants, 0 exact and 0 mechanically admitted
15	char lm tape resegment	character-trigram-constrained-decoding / reverse-tape-dynamic-programming-resegmentation / ordinary-prose-seeds / full-tape-replay / independent-hash-audit	run; 1 exact closure rejected as catalogue/self-unit and 4 non-exact prose proposals
16	constructive seam morph cfg	dependency-tree-seam-csp / agreement-carrying-morphology-transducer / cfg-earley-character-intersection / live-tape-construction / independent-exact-audit	exact witness: Ava saw radar was Ava
17	centerout paired boundary csp verbframe	paired-subject-adjunct-boundary-csp / fresh-verb-frame / ordinary-svo-grammar / live-character-obligation / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
18	free center semantic state machine	free-center-pivot / role-agreement-state / live-bilateral-character-debt / complete-clause-growth / duplicate-sweep-rejection / independent-pointer-sha	diagnostic run; 138-letter intact prose, unresolved character debt
19	reverse lexicon adjacent pp boundary	reverse-lexicon-role-noun-boundary / adjacent-prepositional-attachment / two-boundary-residual-decrease / live-reverse-chart / heldout-scene / independent-	repair run; 0 exact closures and 0 mechanically admitted probes
20	centerout paired boundary csp	paired-subject-adjunct-boundary-csp / ordinary-svo-grammar / live-character-obligation / event-preserving-valency / bounded-single-construction / independe	run; 0 exact closures and 0 mechanically admitted probes
21	reverse lexicon role noun boundary	reverse-lexicon-typed-complement-frame / semantic-role-lattice / object-noun-boundary-selection / outer-residual-reduction / heldout-scene / independent-po	repair run; 0 exact closures and 0 mechanically admitted probes

#	Method / operator	Construction signature	Observed result and next repair
22	centerout grammar boundary repair5	center-out-grammar-state / residual-third-boundary-repair / heldout-scene-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
23	reverse lexicon typed complement frame	reverse-lexicon-centered-complement / typed-transitive-complement-frame / live-reverse-chart-argument-selection / inherited-singular-agreement / heldout-sc	repair run; 0 exact closures and 0 mechanically admitted probes
24	centerout grammar boundary repair4	center-out-grammar-state / next-character-obligation-repair / heldout-semantic-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
25	reverse lexicon centered complement	reverse-lexicon-shared-agreement-seams / centered-complement-clause / inherited-subject-number / three-live-obligations / heldout-scene / independent-point	repair run; 0 exact closures and 0 mechanically admitted probes
26	centerout grammar boundary repair3	center-out-grammar-state / two-boundary-obligation-repair / heldout-time-adjunct-authorship / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
27	reverse lexicon shared agreement seams	reverse-lexicon-boundary-seam-repair / two-lexical-boundaries / shared-singular-agreement-state / cross-seam-residual-closure / heldout-scene / independent	repair run; 0 exact closures and 0 mechanically admitted probes
28	centerout grammar boundary repair2	center-out-grammar-state / outer-letter-directed-adjunct-authorship / heldout-semantic-boundary-repair / event-preserving-valency / independent-exact-hash-	run; 0 exact closures and 0 mechanically admitted probes
29	reverse lexicon boundary seam repair	reverse-lexicon-inflection-repair / joint-determiner-adjective-seam / reverse-obligation-boundary-chart / heldout-scene / independent-pointer-sha-audit	repair run; 0 exact closures and 0 mechanically admitted probes
30	centerout grammar boundary repair	center-out-grammar-state / first-mismatch-boundary-repair / heldout-adjunct-substitution / event-preserving-valency / independent-exact-hash-audit	run; 0 exact closures and 0 mechanically admitted probes
31	semantic slot substitution repair	authored-clause / typed-slot-substitution / pre-render-residual / independent-audit	710 typed clauses and 10,000 bounded pair worlds; 2 exact rows, both punctuation variants of the 38-letter seed; 0 new admissions and 0 reader-eligible rows; next repair adds adjective-noun slots and held-out clauses
32	polar question boundary repair	typed-polar-question / object-fronted-answer / incoming-outgoing-pair-graph / boundary-conditioned-repair	1 exact 44-letter diagnostic row after five targeted complete-answer alternatives, 0 complete-answer closures, 0 mechanically admitted rows, and 0 reader-eligible rows; exact row fails only the proper-palindromic-subspan gate and is fr
33	polar question token boundary csp	typed-polar-question / token-boundary-answer-csp / incoming-outgoing-pair-graph / independent-audit	11,171 lexical states and 3,118 reachable pair states; 1 exact 44-letter diagnostic, 1 non-exact reconstruction retained as a regression fixture, 0 new closures, 0 mechanically admitted rows, and 0 reader-eligible rows; next repair car
34	reversible predicate center extension	targeted-center-insertion / typed-reversible-predicate-answer / independent-pointer-sha-audit	5 authored probes, 5 exact diagnostics at 50-54 letters, 0 mechanically admitted rows, and 0 reader-eligible rows; every closure fails the hidden reversible-span gate and remains fragmentary; next repair replaces the direct pair with a
35	semantic relay svo svo repair	semantic-relay / complete-svo-svo / live-cross-token-debt / no-reversed-token-pairs	8 complete clause pairs, 0 exact closures, 0 mechanically admitted rows, and 0 reader-eligible rows; best frontier is 39 letters with 17 mismatches; next repair adds one authored consequence-object alternative at the recorded mismatch
36	semantic relay outer edge repair	semantic-relay / outer-edge-determiner-object / live-residual / independent-audit	8 targeted authored probes, 0 exact closures, 0 mechanically admitted rows, and 0 reader-eligible rows; best frontier matches 3 outer characters before the actor boundary; next repair co-designs the subject token with the terminal-obje
37	typed reflected residual chart	typed-live-reflected-residual / scene-frame-chart / agreement-valency-before-closure / independent-exact-audit	3 frame pairs, 3 expanded states, 0 exact closures; fresh complete-clause controls 67-68 letters; withheld seed smoke passes independently but is not promoted; next repair changes lexical category only at a live typed boundary

#	Method / operator	Construction signature	Observed result and next repair
38	outer subject lexical pair	complete-agent-np / terminal-locative-suffix / outer-boundary-co-design / independent-two-pointer-sha-audit	256 fresh constructions, 0 exact closures, longest 62 letters; best intact probe 48 letters with 42 mismatches; strict exact and reader gates closed; next: typed preposition/object replacement at the residual seam
39	role labeled grammar csp	role-labeled-subject-verb-object-adjunct / two-clause-csp / live-opposite-edge-residual / independent-sha-audit	4 fresh complete-clause candidates, 73-76 letters, 0 exact closures and 0 strict admissions; best first mismatch is recorded; next: a typed adjunct substitution preserving valency
40	exhaustive pos lattice 44 tape	fixed-diagnostic-tape / exhaustive-dictionary-segmentation / permissive-pos-recognizer / hidden-span-audit	30 dictionary edges but 1 complete segmentation, 0 alternate coherent renderings, 0 strict admissions; internal spans such as anitemmetina close the reader gate; next repair changes an authored outer seam
41	char product automaton	typed-clause-dfa / character-obligation-product / heterogeneous-slot-pairs / center-closure	1,370 character conflicts after 2 surviving first-step states; 0 exact closures and 0 candidate rows; intact controls retained; next: a held-out character-trie phrase inventory with explicit center closure
42	recursive scene grammar	recursive-semantic-scene-grammar / depth-one-derivation-product / independent-two-pointer-sha-audit	729 derivations per side and 531,441 independent pair probes; 0 equation-compatible closures and 0 exact palindromes; next repair carries the residual into recursive valency choices
43	scene lattice semantic valency	human-authored-scene-lattice / typed-valency / contrastive-two-frame / full-residual-propagation	26,244 fresh realizations, 0 exact closures and 0 admitted rows; best intact control is 95 letters but repeated-content checks remain closed; next repair propagates the full mirrored residual vector into each valency choice
44	seed extension wrapper search	internal-seed-grammar-state / independent-wrapper-continuations / live-outer-character-equations	25 fresh wrapper pairs, 0 live survivors, 0 exact closures; longest intact control 103 letters; controls human-unreviewed; next repair authors held-out wrappers ending in the first residual character class
45	reader first clause lattice	human-authored-complete-scenes / typed-center-edge-obligations / reader-first-lattice	10 authored scenes, 400 lattice states, 0 exact closures; longest complete-scene control 67 letters; reader gate closed; catalogue preflight still requires an explicit rerun; next repair broadens center vocabulary and carries residuals
46	heldout character trie phrase inventory product	character-trie-phrase-product / borrowed-regression-controls / provenance-quarantine	3 borrowed exact controls (21, 29, and 51 letters), 0 fresh candidates, 0 admissible rows; quarantined; next repair replaces every control with a fresh held-out POS phrase inventory before counting closures
47	fresh phrase trie product	fresh-authored-pos-inventory / reversed-character-trie-product / typed-complete-clause / independent-audit	232,320 fresh phrase records and 947,222 trie nodes; 232,320 live obligation walks, 0 terminal pairs, 0 exact closures, 0 admissible rows; controls 23-25 letters; next repair adds a center-state grammar transition and held-out semantic
48	center state grammar transition	fresh-authored-center-transition / live-reversed-character-trie / typed-observation-consequence / independent-audits	2,027,520 live trie walks, 48 rendered diagnostic candidates, 0 terminal pairs and 0 exact closures; three disjoint controls; reader gate closed; next repair adds held-out semantic valency agreement across the center transition
49	held out semantic valency frame product	held-out-semantic-valency / typed-subject-verb-object-adjunct / live-mirrored-residual / independent-sha-audit	6 complete frame pairs, 0 exact closures, longest rendered pair 72 letters; independent pointer/SHA audits and provenance pass; reader gate closed; next repair adds a held-out adjunct transition at the center seam
50	fresh clause resegmentation chart	fresh-human-authored-clauses / typed-reverse-tape-chart / epsilon-slot-resegmentation / independent-audit	6 authored clauses, 0 exact candidates and 0 shortcut-gate survivors; best intact control 42 letters with 20 mirrored mismatches; reader gate closed; next repair adds typed plural and inflectional variants at the first residual chart c

5. Reader-facing protocol and next constructive repair

No row has reached the reader study in this snapshot. When one does, the package will contain the intact passage, a word-shuffled control, randomized blinded order, the exact normalization and hash audit, provenance and generator commit, and a reproducible rater form. The immediate next experiment is not another sweep: jointly generate a complete question-answer discourse across the seam, carry agreement and valency state, and reject any candidate containing a proper palindromic subspan before it can be considered for readers.

Reproducibility anchors

Source artifacts: docs/experiment-novelty-registry.json; runs/dream-rsi-dual-boundary-model-authoring-20260918.json; runs/dream-rsi-discourse-ellipsis-20260918.json; runs/dream-rsi-role-live-phrase-lattice-20260918.json; runs/center-state-grammar-transition-20260918.json; runs/held-out-semantic-valency-frame-product-20260918.json; runs/fresh-clause-resegmentation-chart-20260918.json. The report is generated from these files so the table and headline counts can be regenerated after the next Dream-RSI branch.